

ISHIGAKIJIMA, Japan

WMO: 479180

Lat: 24.3367N

Lon: 124.1644E

Elev: 49

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.2	57.0	40.1	36.5	57.7	42.4	40.0	59.1	24.1	66.6	22.3	66.7	12.1	20	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	90.8	80.8	89.9	80.5	89.1	80.3	82.7	87.8	82.1	87.2	81.6	86.8	12.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.3	86.2	80.7	160.0	86.0	80.1	156.5	85.7	46.8	88.2	46.1	87.4	45.5	86.9	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	22.7	20.1	DB	52.8	92.6	2.2	1.1	51.2	93.4	50.0	94.0	48.7	94.6	47.2	95.4	(4)
(5)			WB	46.0	82.9	2.4	1.1	44.3	83.6	42.8	84.2	41.5	84.8	39.7	85.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	66.2	67.1	69.7	74.4	79.0	83.4	85.6	85.1	83.1	79.3	74.8	69.1	(6)
(7)		DBStd	7.78	4.67	5.18	4.85	4.02	3.24	2.82	1.87	1.88	2.45	2.87	3.67	4.59	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	118	44	37	17	1	0	0	0	0	0	0	1	18	(9)
(10)		CDD50	9656	501	480	611	731	900	1002	1104	1087	994	909	745	592	(10)
(11)		CDD65	4299	80	97	163	282	435	552	639	622	544	444	296	145	(11)
(12)		CDH74	40342	71	139	399	1383	3698	6520	8389	8008	6285	3739	1444	267	(12)
(13)		CDH80	13962	0	1	6	94	727	2485	3942	3573	2255	753	123	3	(13)
(14)	Wind	WSAvg	11.2	11.3	11.3	10.8	10.3	9.6	11.2	12.0	11.5	11.2	11.6	11.7	11.8	(14)
(15)	Precipitation	PrecAvg	81.6	5.0	4.6	5.1	5.7	8.1	7.9	6.4	9.5	9.6	7.6	6.4	5.5	(15)
(16)		PrecMax	132.8	14.1	22.8	14.8	18.9	25.1	27.1	24.8	24.3	32.5	36.8	17.6	18.2	(16)
(17)		PrecMin	43.9	1.3	0.5	0.9	0.2	0.2	0.5	0.3	1.0	1.2	0.7	0.7	0.8	(17)
(18)		PrecStd	17.9	2.8	3.3	3.1	4.5	5.7	5.3	5.4	5.5	6.3	6.9	3.7	3.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.0	79.2	80.9	84.0	87.9	90.7	91.9	92.0	90.8	88.4	84.7	80.3	(19)
(20)			MCWB	70.3	72.8	74.6	77.3	80.2	82.0	81.2	80.6	80.2	79.4	76.8	72.7	(20)
(21)		2%	DB	75.9	77.1	79.1	82.4	86.1	89.3	90.8	90.7	89.5	86.5	82.9	78.3	(21)
(22)			MCWB	68.8	70.8	72.9	76.1	79.0	81.3	80.9	80.5	79.7	77.9	75.4	71.0	(22)
(23)		5%	DB	74.4	75.7	77.7	81.0	84.8	88.2	89.9	89.7	88.4	85.1	81.4	76.9	(23)
(24)			MCWB	67.7	69.8	71.7	75.4	78.2	80.9	80.6	80.3	79.1	76.8	74.1	70.1	(24)
(25)		10%	DB	72.9	74.4	76.3	79.7	83.7	87.3	89.1	88.7	87.3	83.8	80.0	75.4	(25)
(26)			MCWB	66.5	68.6	70.7	74.4	77.5	80.5	80.4	80.0	78.6	75.9	73.1	68.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.5	74.2	75.8	78.9	81.7	83.4	83.5	83.5	82.4	81.0	78.3	75.2	(27)
(28)			MCDWB	75.6	77.8	79.3	82.4	85.9	88.3	89.2	88.2	88.3	85.7	82.7	78.2	(28)
(29)		2%	WB	70.8	72.6	74.5	77.6	80.5	82.6	82.6	82.5	81.4	79.9	77.2	73.2	(29)
(30)			MCDWB	74.2	76.0	77.9	80.9	84.3	87.1	88.1	87.7	87.1	84.6	81.2	76.8	(30)
(31)		5%	WB	69.2	71.2	73.2	76.6	79.5	81.9	81.9	81.8	80.7	78.7	76.0	71.6	(31)
(32)			MCDWB	73.3	74.8	76.9	79.9	83.3	86.4	87.5	87.2	86.1	83.6	80.1	75.5	(32)
(33)		10%	WB	67.6	69.6	71.9	75.5	78.7	81.4	81.4	81.0	79.9	77.6	74.8	69.9	(33)
(34)			MCDWB	72.1	73.5	75.7	78.8	82.5	86.0	87.2	86.6	85.4	82.5	79.0	74.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	7.7	8.1	7.5	7.4	6.6	6.9	7.2	7.8	7.7	7.4	7.3	(35)
(36)			MCDBR	8.7	8.4	8.4	7.5	7.6	6.6	7.4	8.1	8.4	8.4	8.1	8.3	(36)
(37)			MCWBR	5.6	5.8	6.0	4.6	4.1	3.0	3.0	3.3	3.6	3.9	4.5	5.5	(37)
(38)		5% WB	MCDBR	7.8	7.9	8.0	6.9	7.0	6.4	6.9	7.3	7.6	7.7	7.3	7.4	(38)
(39)			MCWBR	6.1	6.4	6.9	4.9	4.2	3.3	3.3	3.4	3.7	3.9	4.8	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.488	0.507	0.595	0.600	0.522	0.486	0.481	0.499	0.495	0.499	0.491	0.495	(40)	
(41)		taud	2.074	2.052	1.850	1.891	2.160	2.298	2.309	2.245	2.228	2.161	2.151	2.062	(41)	
(42)		Ebn at Noon	240	246	231	233	251	258	259	255	254	246	238	232	(42)	
(43)		Edh at Noon	47	51	65	64	49	42	42	44	44	45	43	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	752	961	1163	1376	1626	1835	2040	1845	1636	1302	932	736	(44)	
(45)		RadStd	118	172	148	177	192	145	133	135	177	141	103	100	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.47	N/A	N/A	+0.59	+1.04	+1.24	N/A	N/A	+176	+162	(46)
(47)	Regional (3 neighbors)	+0.41	N/A	N/A	+0.56	+0.77	+0.83	N/A	N/A	+135	+124	(47)

Nomenclature: See separate page